

GESTURE RECOGNITION USING NI LABVIEW

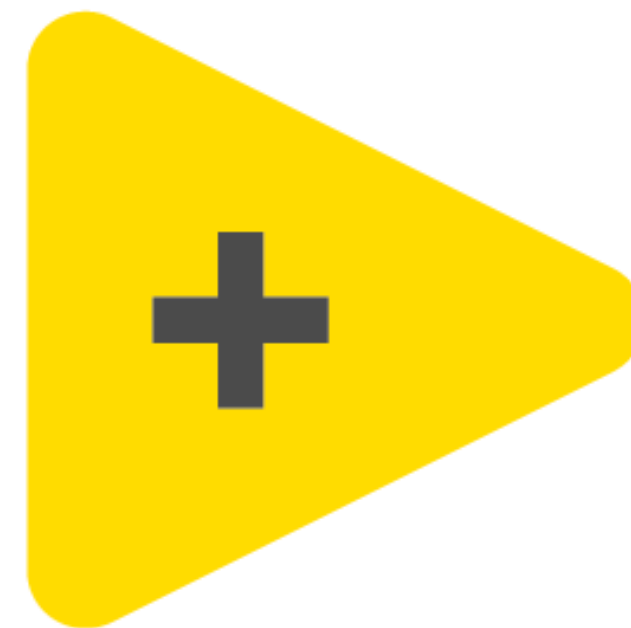
A Vision-Based Approach for Human-
Computer Interaction

INTRODUCTION

- Gesture recognition enables computers to interpret human gestures as commands.
- Enhances human-computer interaction (HCI) without physical touch.
- Applications: AR/VR, gaming, robotics, accessibility, automation.

WHY USE NI LABVIEW?

- Graphical programming simplifies development.
- Built-in vision tools for image processing.
- Real-time processing capabilities.
- Supports seamless hardware integration with cameras and sensors.



LabVIEW™

NI VISION DEVELOPMENT MODULE (VDM)

- Provides tools for:
- Image acquisition (NI-IMAQdx drivers)
- Image preprocessing (filtering, edge detection)
- Feature extraction (contours, blobs, shape patterns)
- Pattern matching and recognition.

SYSTEM COMPONENTS

- Software: NI LabVIEW, NI Vision Development Module, NI-IMAQdx drivers.
- Hardware: Camera (USB/IP), NI-compatible hardware.
- Optional: External lighting for better accuracy.

WORKFLOW OVERVIEW

- Image Acquisition
- Image Preprocessing
- Feature Extraction
- Gesture Recognition
- Output and Application

IMAGE ACQUISITION

- Uses NI-IMAQdx for interfacing with cameras.
- Supports USB3 Vision, GigE Vision, and DirectShow cameras.
- Live video feed integration using NI MAX & LabVIEW.

IMAGE PREPROCESSING

- Convert RGB to grayscale for faster processing.
- Apply filters (Gaussian, Median) to reduce noise.
- Use thresholding to segment hand from background.
- Edge detection (Canny, Sobel) for feature extraction.

FEATURE EXTRACTION

- Contour detection for hand shape analysis.
- Blob analysis to track hand movement.
- Convex hull & defect identification for finger counting.

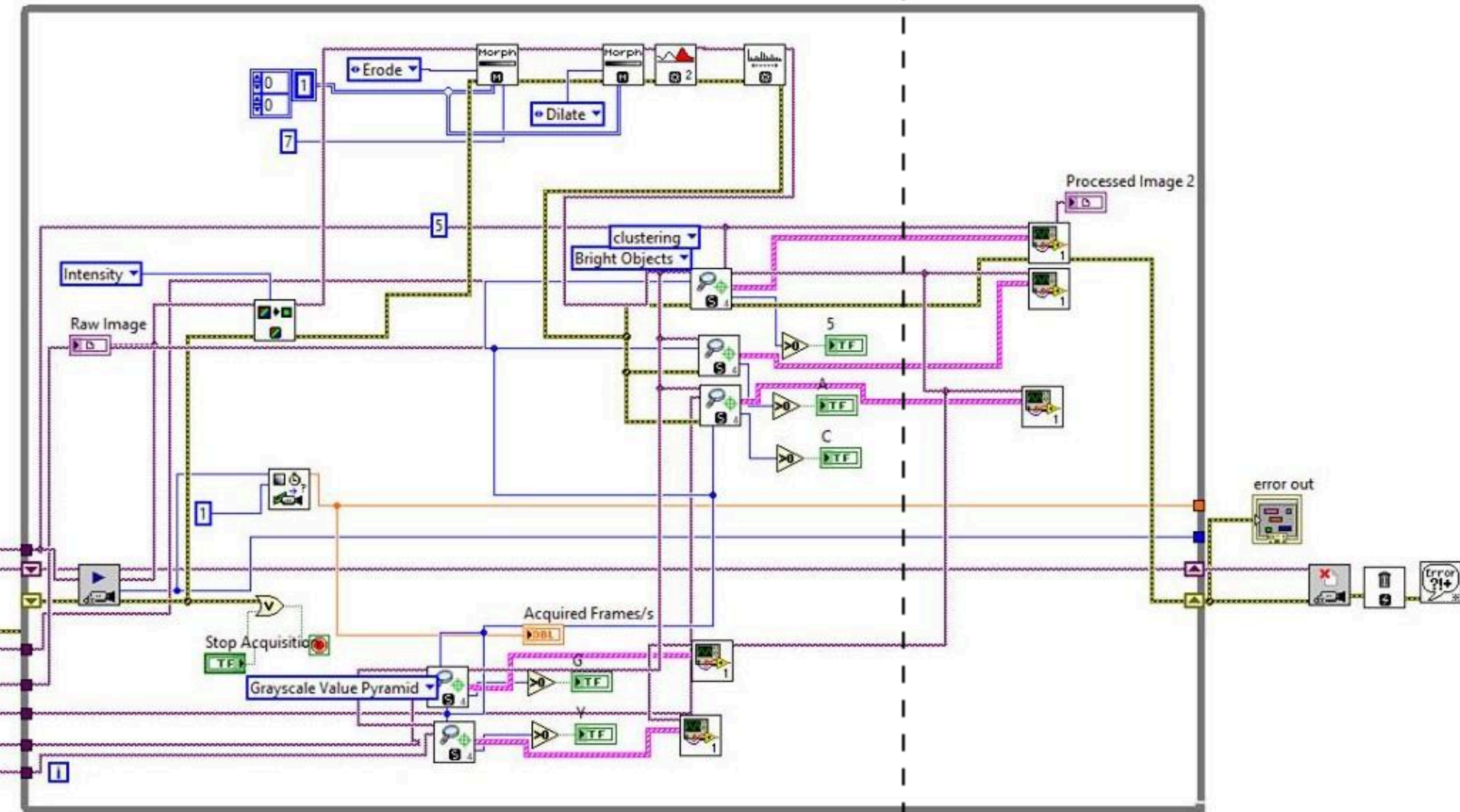
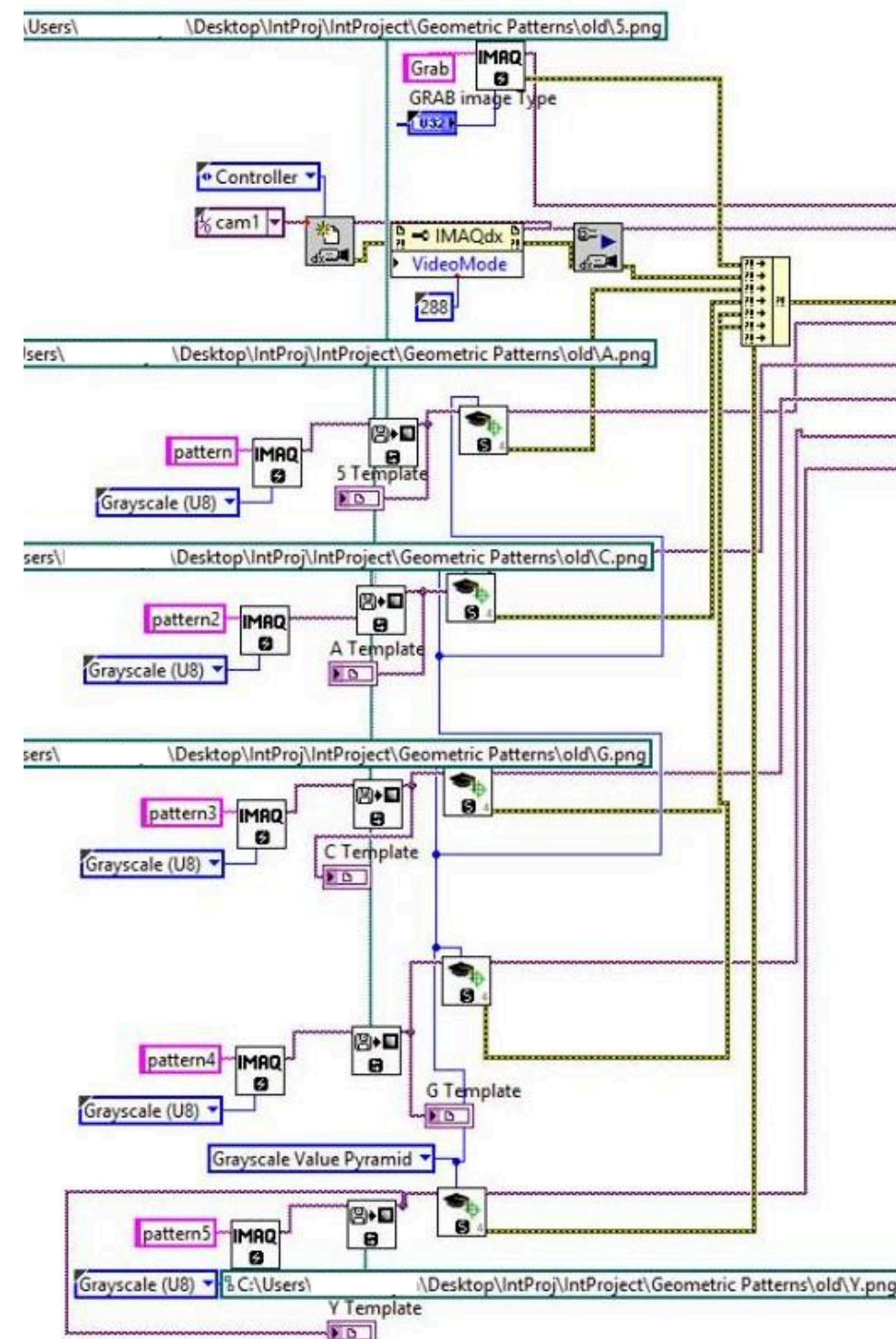
GESTURE RECOGNITION ALGORITHM

- Extract features from processed images.
- Compare with predefined gesture templates.
- Apply pattern matching or machine learning.
- Decision logic determines gesture classification.

ACQUIRE

ANALYZE

PRESENT



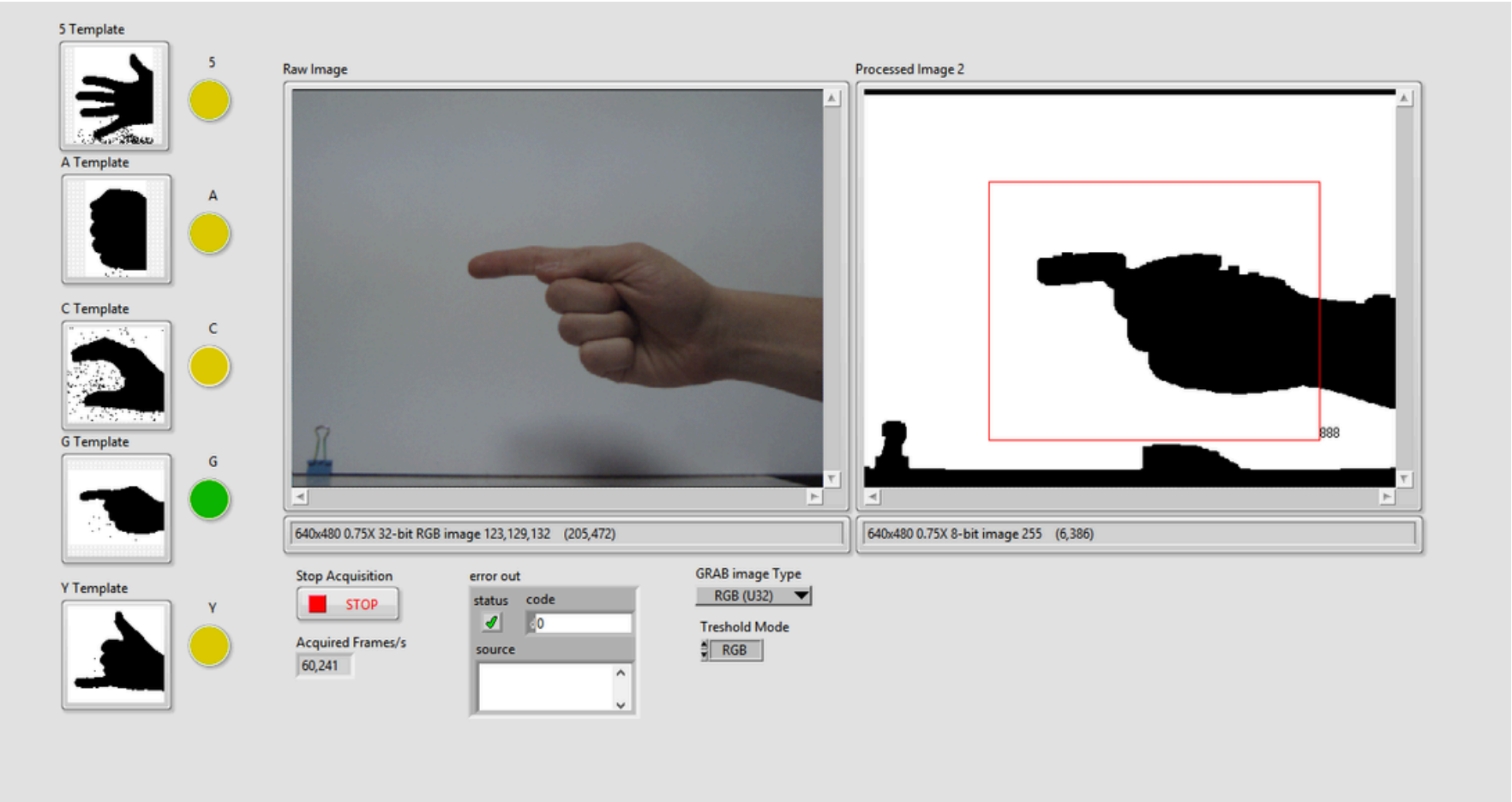
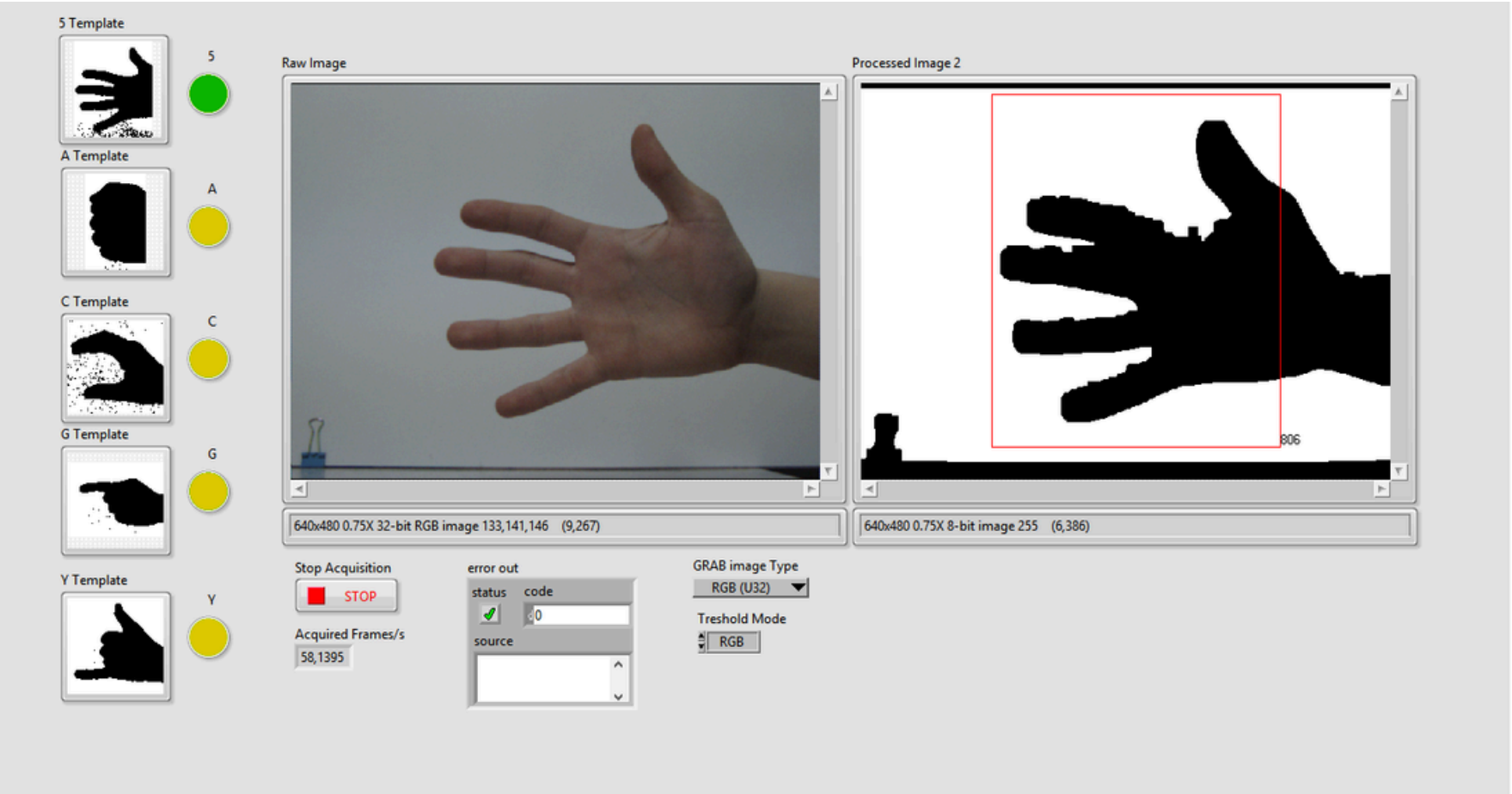
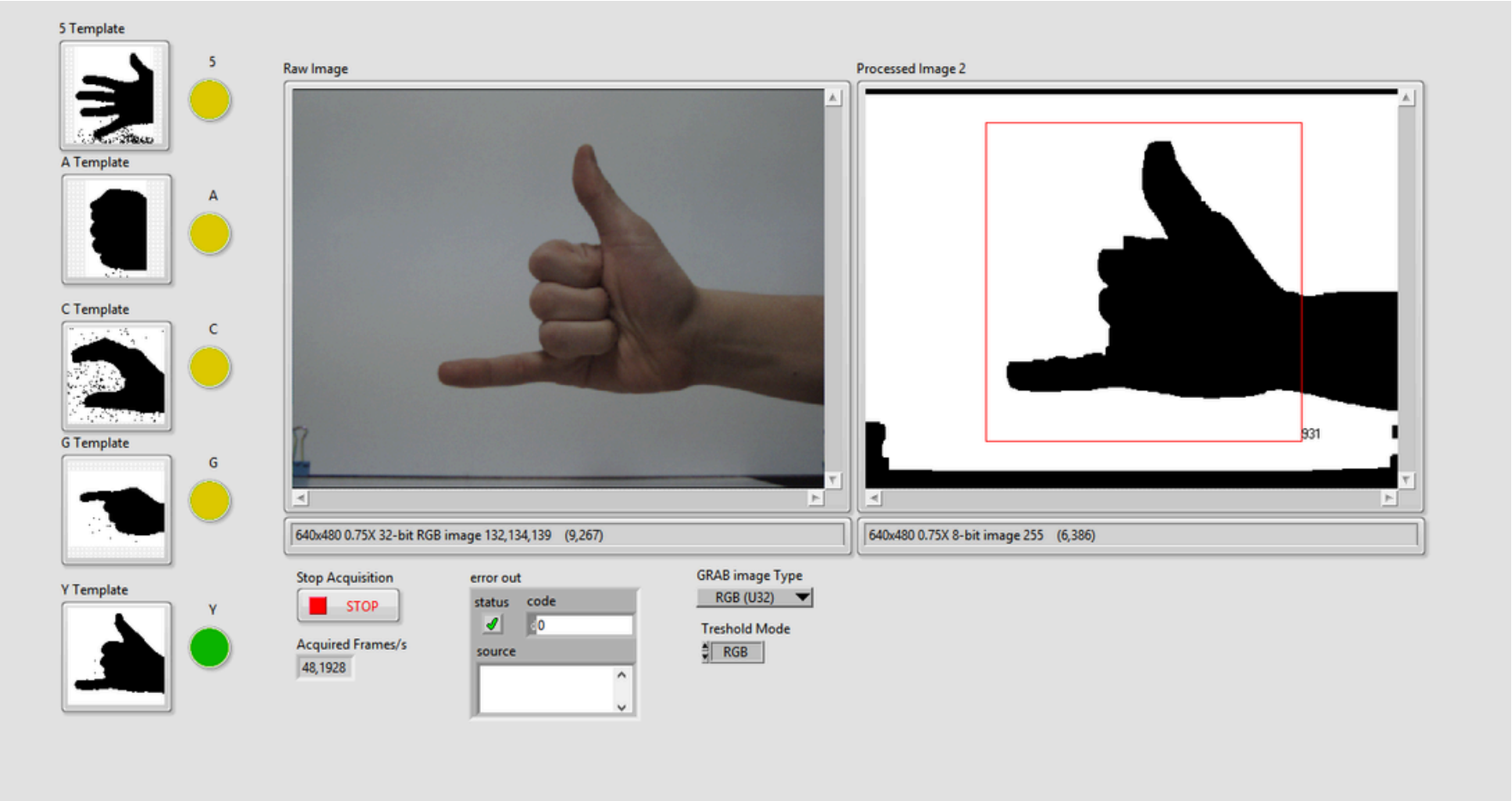
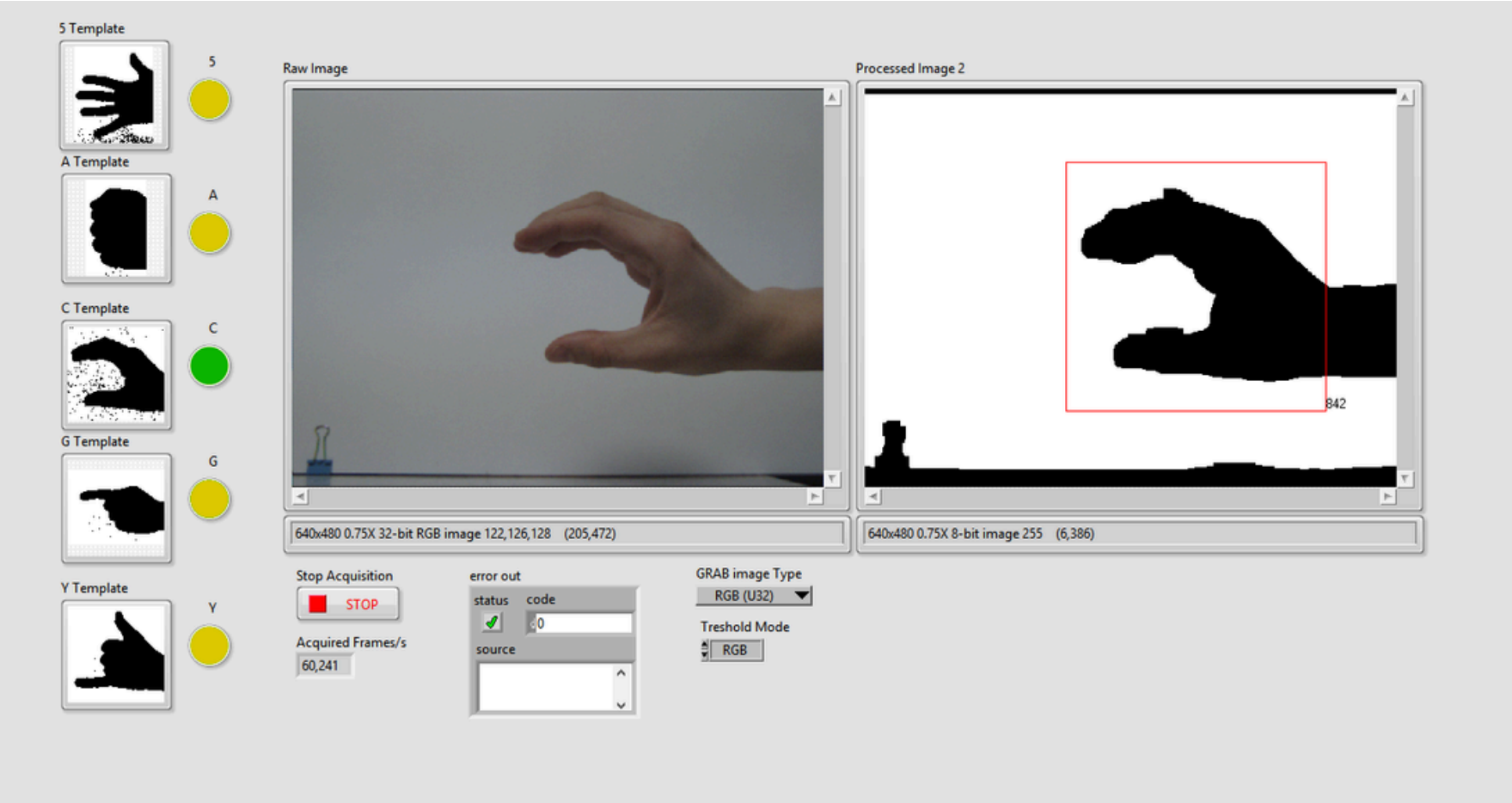
BLOCKS USED FROM LABVIEW

- IMAQdx Open Camera – Used for interfacing with the camera and acquiring images.
- IMAQdx Grab – Captures live image frames from the camera.
- IMAQ Create – Creates an image buffer for processing.
- IMAQ Read File – Loads image files from disk into memory.
- IMAQ Grayscale (U8) Conversion – Converts images to grayscale format.
- IMAQ Image Template Matching – Performs pattern matching against stored templates.
- IMAQ Value Pyramid – Used for multiresolution analysis.

BLOCKS USED FROM LABVIEW

- IMAQ Erode – Performs morphological erosion operation.
- IMAQ Dilate – Performs morphological dilation operation.
- IMAQ Clustering Bright Objects – Used for object segmentation based on brightness.
- IMAQ Display – Outputs processed images.
- IMAQ Stop Acquisition – Halts image acquisition.
- IMAQ Error Out – Handles errors during processing.

These components work together in an Acquire → Analyze → Present pipeline for image processing and analysis in LabVIEW.



GRAPHICAL USER INTERFACE (GUI)

- Displays raw and processed images.
- Indicators for detected gestures.
- User controls for image acquisition and threshold adjustment.

CHALLENGES & LIMITATIONS

- Environmental factors:
Background noise, lighting conditions.
- Real-time processing constraints.
- Hand shape variability affecting accuracy.

FUTURE IMPROVEMENTS

- AI/ML integration for better recognition accuracy.
- Gesture recognition using 3D sensors.
- Improved preprocessing algorithms (adaptive thresholding, advanced edge detection).



THANK YOU

